Achieve tight oracle complexity for composite nonconvex optimization with affine constraints

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Website #14 | Solver verdict: PARTIAL | Verifier verdict: n/a | Shown: solver output

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Problem

We are given the affinely constrained composite problem

$$\min_{x\in\mathbb{R}^d} F_0(x) := f_0(x) + g(x) \qquad \text{s.t.} \qquad Ax + b = 0,$$

where f_0 has L_f-Lipschitz gradient and

$$g(x) = \bar{g}(\bar{A}x + \bar{b}),$$

with $\bar{g}$ proper, lower semicontinuous, convex, $\operatorname{relint}(\operatorname{dom}\bar{g}) \neq \emptyset$, and $\operatorname{dom}\bar{g}$ not a singleton. An ε-stationary point is an x for which there exists $\gamma \in \mathbb{R}^n$ such that

$$\max\Big\{\operatorname{dist}\big(0, \nabla f_0(x) + A^\top\gamma + \partial g(x)\big),\ \|Ax + b\|\Big\} \le \varepsilon.$$

The question asks whether there exists a first-order method using only the Class-1 oracle

$$(x, z, \eta) \mapsto \big(\nabla f_0(x),\, Ax,\, A^\top z,\, \operatorname{prox}_{\eta g}(x)\big)$$

that, for *every* instance in the stated class, returns such an ε-stationary point within

$$\tilde{O}\big(\kappa([\bar{A}; A])\, L_f\, \Delta F_0\, \varepsilon^{-2}\big)$$

oracle calls, matching the proved lower bound up to logarithms.

Alternative readings considered

There are two readings.

1. *Literal reading:* the claim quantifies over *all* syntactically admissible instances exactly as written, with no hidden feasibility or constraint-qualification assumptions. Under this reading the statement is false, because some admissible instances have no ε-stationary point at all for small ε.

2. *Natural optimization reading:* one restricts to the meaningful subclass where the problem is feasible and the initialization/regularity assumptions needed for KKT-based stationarity are in force. For this repaired reading there is now an affirmative answer in the literature: a 2025 follow-up paper of Liu–Lin–Xu gives a Class-1 method closing the gap left open by the 2025 lower-bound paper. (ar5iv.org)

I treat the literal reading first, because the prompt asked for the hardest logically exact reading; then I prove the strongest audited positive result I can for the natural repaired reading.

Related literature found

1. **Wei Liu, Qihang Lin, Yangyang Xu (2025), "Lower Complexity Bounds of First-order Methods for Affinely Constrained Composite Non-convex Problems."** Classification: `BACKGROUND/PARTIAL`. This is the lower-bound paper underlying the prompt. It defines the Class-1 oracle, proves the $\Omega(\kappa([\bar{A}; A])L_f\Delta F_0\varepsilon^{-2})$ lower bound, and explicitly says tightness for Class 1 was unresolved there. (ar5iv.org)

2. **Wei Liu, Qihang Lin, Yangyang Xu (2025), "A Near-optimal Method for Linearly Constrained Composite Non-convex Non-smooth Problems."** Classification: `SOLVES` (for the repaired reading audited below). This paper states that the earlier Class-1 gap is closed by a PG-RPD method using the same oracle model; its abstract says the oracle complexity matches the lower bounds up to logarithmic factors, and Corollary 3.8 concerns the original problem (P), not only the splitting reformulation. (arxiv.org)

3. **Wei Liu, Qihang Lin, Yangyang Xu (2023), "First-order Methods for Affinely Constrained Composite Non-convex Non-smooth Problems: Lower Complexity Bound and Near-optimal Methods."** Classification: `PARTIAL`. This older arXiv version already had a near-optimal method, but explicitly for the *more relaxed* oracle/model (the splitting/Class-2 side), so it does not by itself settle the present Class-1 question. (arxiv.org)

Proof sketch

There are two logically distinct tasks.

First, the literal universal statement is too strong: if the affine constraint is infeasible, then no point can have small feasibility residual, so no algorithm can possibly succeed on *every* instance. A one-dimensional example already disproves the statement exactly as written.

Second, the mathematically interesting question is whether the 2025 Class-1 lower bound is tight on the intended feasible/KKT-admitting class. Here the key observation is that redundant rows in A and redundant rows in the representation matrix $\bar{A}$ can be removed by thin SVD compression without changing:

- the objective F_0,
- the feasible set,
- the stationarity notion,
- the Class-1 oracle model up to one-to-one simulation, or
- the joint condition number $\kappa([\bar{A}; A])$.

So one may reduce to a full-row-rank representation.

After that reduction, the 2025 PG-RPD theorem of Liu–Lin–Xu applies on the repaired subclass satisfying their additional regularity assumptions. Their paper explicitly says it uses the same Class-1 oracle and that its complexity matches the earlier lower bound up to logarithmic factors. Pulling the reduced multipliers back to the original coordinates gives the desired ε-stationarity statement for the original instance.

The only imported part is the actual PG-RPD convergence/complexity theorem; everything else below (counterexample, rank-compression lemmas, condition-number preservation, oracle simulation, and transfer of stationarity) is proved directly here.

Main result

Theorem 1. *Let $M := [\bar{A}; A]$.*

(i) Under the literal reading of the prompt, the answer is no*: there is no algorithm, first-order or otherwise, that can succeed for every syntactically admissible instance.*

(ii) Consider instead the repaired subclass consisting of instances for which the initial point $x^{(0)}$ is feasible and satisfies the additional regularity assumptions needed by the 2025 PG-RPD theorem of Liu–Lin–Xu after the full-row-rank compressions constructed in Lemmas 2 and 3 (in particular, the reduced instance has full-row-rank constraint matrices and a feasible relative-interior initialization; see the hypothesis audit in Section Invoked results*). On this repaired subclass there exists a first-order method in Algorithm Class 1 that returns an ε-stationary point using*

$$\tilde{O}\big(\kappa(M)\, L_f\, \Delta F_0\, \varepsilon^{-2}\big)$$

oracle calls.

Proof

Lemma 1. *The literal universally quantified statement in the prompt is false.*

Proof. Take $d = n = \bar{n} = 1$, let

$$f_0(x) \equiv 0, \qquad A = [0], \qquad b = [1], \qquad \bar{A} = [1], \qquad \bar{b} = [0],$$

and let $\bar{g} : \mathbb{R} \to \mathbb{R}$ be $\bar{g}(t) \equiv 0$. Then $\bar{g}$ is proper, l.s.c., convex, $\operatorname{dom} \bar{g} = \mathbb{R}$ has nonempty relative interior, and $\operatorname{dom} \bar{g}$ is not a singleton. Also

$$g(x) = \bar{g}(\bar{A}x + \bar{b}) = 0 \quad \forall x,$$

so this is indeed an instance of the stated problem class.

For every $x \in \mathbb{R}$,

$$Ax + b = 1,$$

hence $\|Ax + b\| = 1$. Therefore, for every $\varepsilon < 1$,

$$\max\Big\{\operatorname{dist}\big(0, \nabla f_0(x) + A^\top \gamma + \partial g(x)\big),\ \|Ax + b\|\Big\} \geq 1 > \varepsilon$$

for all x and all γ. So no ε-stationary point exists at all. Consequently no algorithm can guarantee success on *every* instance as literally stated. □

The rest of the proof addresses the natural repaired reading.

Lemma 2. *Assume the affine system $Ax + b = 0$ is feasible. Let $r = \operatorname{rank}(A)$, and let*

$$A = U_A \Sigma_A V_A^\top$$

be a thin SVD, with $U_A \in \mathbb{R}^{n\times r}$, $\Sigma_A \in \mathbb{R}^{r\times r}$ diagonal positive, and $V_A \in \mathbb{R}^{d\times r}$. Define

$$A_r := \Sigma_A V_A^\top \in \mathbb{R}^{r\times d}, \qquad b_r := U_A^\top b \in \mathbb{R}^r.$$

Then:

(a) *A_r has full row rank and $A_r^\top A_r = A^\top A$.*

(b) *For every $x \in \mathbb{R}^d$,*

$$Ax + b = U_A(A_r x + b_r).$$

In particular,

$$Ax + b = 0 \iff A_r x + b_r = 0, \qquad \|Ax + b\| = \|A_r x + b_r\|.$$

(c) *For every $\lambda \in \mathbb{R}^r$,*

$$A_r^\top \lambda = A^\top (U_A \lambda).$$

Hence

$$\operatorname{dist}\big(0, \nabla f_0(x) + A_r^\top \lambda + \partial g(x)\big) = \operatorname{dist}\big(0, \nabla f_0(x) + A^\top (U_A \lambda) + \partial g(x)\big).$$

(d) *One oracle query (x, λ, η) for the reduced matrix A_r can be simulated by one oracle query $(x, U_A\lambda, \eta)$ for the original matrix A, followed by the post-processing $A_r x = U_A^\top (Ax)$.*

Proof. Because $Ax + b = 0$ is feasible, there exists x_f with $b = -Ax_f \in \operatorname{range}(A) = \operatorname{range}(U_A)$. Hence $b = U_A U_A^\top b$.

Part (a): A_r has rank r, so it has full row rank; and

$$A_r^\top A_r = V_A \Sigma_A^\top \Sigma_A V_A^\top = A^\top A.$$

Part (b):

$$U_A(A_r x + b_r) = U_A \Sigma_A V_A^\top x + U_A U_A^\top b = Ax + b.$$

Since the columns of U_A are orthonormal, $\|U_A y\| = \|y\|$ for every $y \in \mathbb{R}^r$. The claimed equivalence and equality of norms follow.

Part (c):

$$A^\top (U_A \lambda) = V_A \Sigma_A U_A^\top U_A \lambda = V_A \Sigma_A \lambda = A_r^\top \lambda.$$

The distance identity is immediate.

Part (d) is exactly the previous identities rewritten operationally. □

Lemma 3. *Let $\bar{r} = \operatorname{rank}(\bar{A})$, and let*

$$\bar{A} = U_{\bar{A}} \Sigma_{\bar{A}} V_{\bar{A}}^\top$$

be a thin SVD. Extend $U_{\bar{A}}$ to an orthogonal matrix $[U_{\bar{A}}, U_{\bar{A},\perp}]$, and decompose

$$\bar{b} = U_{\bar{A}} q + U_{\bar{A},\perp} c$$

with $q \in \mathbb{R}^{\bar{r}}$. Define

$$\bar{A}_r := \Sigma_{\bar{A}} V_{\bar{A}}^\top, \qquad \bar{b}_r := q,$$

and define a new convex function $\bar{g}_r : \mathbb{R}^{\bar{r}} \to \mathbb{R} \cup \{+\infty\}$ by

$$\bar{g}_r(y) := \bar{g}(U_{\bar{A}} y + U_{\bar{A},\perp} c).$$

Then:

(a) *$\bar{A}_r$ has full row rank and $\bar{A}_r^\top \bar{A}_r = \bar{A}^\top \bar{A}$.*

(b) The original regularizer is represented exactly as

$$g(x) = \bar{g}(\bar{A}x + \bar{b}) = \bar{g}_r(\bar{A}_r x + \bar{b}_r) \qquad \forall x \in \mathbb{R}^d.$$

Hence g itself, its proximal mapping, and its convex subdifferential are unchanged.

(c) If $x^{(0)}$ satisfies $\bar{A}x^{(0)} + \bar{b} \in \operatorname{relint}(\operatorname{dom}\bar{g})$, then

$$\bar{A}_r x^{(0)} + \bar{b}_r \in \operatorname{relint}(\operatorname{dom}\bar{g}_r).$$

Proof. Part (a) is the same SVD calculation as in Lemma 2.

For part (b), compute

$$U_{\bar{A}}(\bar{A}_r x + \bar{b}_r) + U_{\bar{A},\perp}c = U_{\bar{A}}\Sigma_{\bar{A}}V_{\bar{A}}^\top x + U_{\bar{A}}q + U_{\bar{A},\perp}c = \bar{A}x + \bar{b}.$$

Applying $\bar{g}$ to both sides yields

$$\bar{g}_r(\bar{A}_r x + \bar{b}_r) = \bar{g}(\bar{A}x + \bar{b}) = g(x).$$

Therefore the pointwise function g is unchanged, so $\operatorname{prox}_{\eta g}$ and $\partial g(x)$ are unchanged as well.

For part (c), let

$$y_0 := \bar{A}x^{(0)} + \bar{b}, \qquad w_0 := \bar{A}_r x^{(0)} + \bar{b}_r,$$

and define the affine bijection

$$T : \mathbb{R}^{\bar{r}} \to S := U_{\bar{A}}\mathbb{R}^{\bar{r}} + U_{\bar{A},\perp}c, \qquad T(w) = U_{\bar{A}}w + U_{\bar{A},\perp}c.$$

Then $T(w_0) = y_0$, and

$$\operatorname{dom}\bar{g}_r = T^{-1}(\operatorname{dom}\bar{g} \cap S).$$

Since $y_0 \in \operatorname{relint}(\operatorname{dom}\bar{g})$, by definition there exists $\rho > 0$ such that the relative ball

$$B_{\operatorname{aff}(\operatorname{dom}\bar{g})}(y_0, \rho) \subset \operatorname{dom}\bar{g}.$$

Intersecting with S gives

$$B_{\operatorname{aff}(\operatorname{dom}\bar{g})\cap S}(y_0, \rho) \subset \operatorname{dom}\bar{g} \cap S.$$

Now T is an isometry from $\mathbb{R}^{\bar{r}}$ onto S, so T^{-1} maps the last relative ball to a relative ball around w_0 contained in $T^{-1}(\operatorname{dom}\bar{g} \cap S) = \operatorname{dom}\bar{g}_r$. Therefore $w_0 \in \operatorname{relint}(\operatorname{dom}\bar{g}_r)$. □

Lemma 4. *Let*

$$M = [\bar{A}; A], \qquad M_r = [\bar{A}_r; A_r].$$

Then

$$M_r^\top M_r = M^\top M.$$

Consequently M and M_r have the same nonzero singular values, and hence

$$\kappa(M_r) = \kappa(M).$$

Proof. By Lemmas 2 and 3,

$$M_r^\top M_r = \bar{A}_r^\top \bar{A}_r + A_r^\top A_r = \bar{A}^\top \bar{A} + A^\top A = M^\top M.$$

The nonzero eigenvalues of $M^\top M$ are the squares of the nonzero singular values of M, and similarly for M_r. Hence the nonzero singular values coincide, so $\kappa(M_r) = \kappa(M)$. □

Proof of the theorem. Part (i) is Lemma 1.

For part (ii), apply Lemmas 2 and 3 to obtain an equivalent reduced instance with full-row-rank matrices $A_r, \bar{A}_r$. By Lemma 3(b), the objective F_0 is unchanged pointwise, so the same starting point $x^{(0)}$ yields the same quantity

$$\Delta F_0 = F_0(x^{(0)}) - \inf_x F_0(x).$$

By Lemma 2(b), the feasible set is unchanged. By Lemma 3(c), the feasible relative-interior initialization required in the imported PG-RPD theorem is preserved under the representation change. By Lemma 4, the joint condition number is exactly preserved.

Now invoke the 2025 PG-RPD theorem of Liu–Lin–Xu on the reduced instance. Their paper explicitly takes as oracle the same Class-1 oracle from the lower-bound paper, says that its goal is to close the Class-1 gap left there, and states in the abstract that the resulting oracle complexity matches the earlier lower bounds up to logarithmic factors; moreover Corollary 3.8 is a guarantee for the original problem (P), not only the splitting reformulation. (arxiv.org) Therefore, on every reduced instance satisfying their audited assumptions, there exists a Class-1 method producing an ε-stationary (their terminology: ε-KKT) point within

$$\tilde{O}\big(\kappa(M_r)\, L_f\, \Delta F_0\, \varepsilon^{-2}\big) = \tilde{O}\big(\kappa(M)\, L_f\, \Delta F_0\, \varepsilon^{-2}\big)$$

oracle calls, where the equality is Lemma 4. (arxiv.org)

Finally, translate the reduced output back to the original coordinates. If the reduced method returns x and a reduced multiplier λ, define

$$\gamma := U_A \lambda.$$

By Lemma 2(b)–(c),

$$\|Ax + b\| = \|A_r x + b_r\|$$

and

$$\operatorname{dist}\big(0, \nabla f_0(x) + A^\top \gamma + \partial g(x)\big) = \operatorname{dist}\big(0, \nabla f_0(x) + A_r^\top \lambda + \partial g(x)\big).$$

Hence any ε-stationary point for the reduced instance is also an ε-stationary point for the original instance. By Lemma 2(d), each reduced oracle query is simulated by one original oracle query, so the oracle complexity bound is preserved up to a constant factor. This proves part (ii). □

Gap to the full statement

The literal statement is false, so in that strict sense there is no gap: the answer is negative.

For the *interesting repaired reading*, however, my affirmative theorem is weaker than the raw prompt. The remaining uncovered cases are the prompt-instances that do *not* satisfy the additional regularity/initialization hypotheses of the imported 2025 PG-RPD theorem after reduction. Concretely, I have *not* proved the near-optimal Class-1 upper bound for:

- feasible instances where the designated initial point $x^{(0)}$ is not feasible;
- feasible instances where every feasible point lies on the relative boundary of $\operatorname{dom} \bar{g}$;
- any prompt-instance failing the additional regularity assumption appearing as Assumption 1(e) in Liu–Lin–Xu (2025).

So the strongest rigorously audited conclusion I can defend is: *the prompt as written is false, but the intended Class-1 tightness question has an affirmative answer on the natural reduced subclass covered by the 2025 PG-RPD theorem.*

Self red-team

Counterexample search. I first tried to break the literal claim with the smallest possible infeasible example. The 1-dimensional instance in Lemma 1 already works. I also checked that $\bar{g} \equiv 0$ keeps all prompt hypotheses valid, and by taking $\bar{A} = 1$ rather than $\bar{A} = 0$ I avoid the degenerate case $M = 0$.

Plausible referee objection 1: "Your SVD compression may change the condition number $\kappa([\bar{A}; A])$." Answer: Lemma 4 proves the stronger identity $M_r^\top M_r = M^\top M$, so all nonzero singular values are preserved exactly.

Plausible referee objection 2: "Changing the representation of g may change the subdifferential and hence stationarity." Answer: Lemma 3(b) shows the pointwise function g is unchanged, so $\partial g(x)$ and $\mathrm{prox}_{\eta g}$ are literally unchanged.

Plausible referee objection 3: "The imported theorem may solve only the splitting problem, not the original one." Answer: the 2025 PG-RPD paper explicitly frames itself as closing the gap for the original Class-1 oracle, and Corollary 3.8 is stated for the original problem (P). (arxiv.org)

Boundary checks.

- If $A = 0$ and $b \neq 0$, infeasibility makes the literal statement impossible; this is exactly the mechanism used in Lemma 1.
- If $A = 0$ and $b = 0$, Lemma 2 reduces to the trivial empty-row system.
- If $\bar{A} = 0$, Lemma 3 reduces g to a constant-composition case; the proof still works.
- I checked that my reduction does *not* accidentally prove a claim known to fail: the unresolved part I leave unresolved is precisely the non-Slater / nonqualified region.

Ideas tried

1. **Direct simulation of the 2023 Class-2 IPG by the Class-1 oracle.** Idea: start from the 2023 splitting/IPG upper bound and try to express each split-oracle operation through $\nabla f_0(x), Ax, A^\top z, \mathrm{prox}_{\eta g}(x)$. Obstruction: the split method works in a different oracle/model; a black-box constant-overhead reduction from the split proximal operations to the original Class-1 oracle is not available from the prompt data alone. This route did not yield a rigorous proof.

2. **Rank-compress the data and import the 2025 PG-RPD theorem.** Idea: remove redundant rows from A and from the representation matrix of g, while preserving F_0, stationarity, the oracle, and κ. Outcome: success on the qualified subclass; this became the positive half of the main theorem.

3. **Eliminate the feasible/relative-interior initialization assumption.** Idea: project an arbitrary initial point onto the affine set using explicit linear algebra, then repair domain feasibility by a proximal step. Obstruction: projection need not preserve membership in $\mathrm{dom}\, g$, and I found no uniform argument bounding the new initial gap by the prompt's ΔF_0. So I could not extend the audited positive theorem to arbitrary starts.

4. **Try to disprove the meaningful feasible case as well.** Idea: build a feasible boundary-only instance, e.g. an indicator regularizer whose feasible set touches only the relative boundary, and show no near-optimal Class-1 algorithm can exist. Obstruction: I found no such

lower-bound argument. The simple counterexamples I could certify were all infeasible ones, which only disprove the literal reading.

Invoked results

1. **Liu–Lin–Xu (2025), lower-bound paper.** Classification: `BACKGROUND/PARTIAL`. *Statement used.* Their Problem Class 1 uses the same Class-1 oracle as in the prompt, and they prove a lower bound of order

$$\Omega\big(\kappa([\bar{A}; A])\, L_f\, \Delta F_0\, \varepsilon^{-2}\big)$$

   for Class 1; they also state that tightness for Class 1 was unresolved there. (ar5iv.org) *Hypothesis audit.* This is exactly the oracle model and lower-bound quantity appearing in the prompt. I use this result only to identify the target rate and to interpret the phrase "matches the lower bounds up to logarithmic factors" in the follow-up paper.

2. **Liu–Lin–Xu (2025), PG-RPD follow-up paper.** Classification: `SOLVES` (imported external proof). *Statement used.* The paper considers the same Class-1 oracle, proposes Algorithm 1 (PG-RPD), states in the abstract that its oracle complexity for the original problem matches the previous lower bounds up to logarithmic factors, and Corollary 3.8 gives an ε-KKT guarantee for the original problem (P). (arxiv.org) *Hypothesis audit.*

   - Same oracle model: yes; their equation (1.1) is exactly the Class-1 oracle inherited from the lower-bound setup. (ar5iv.org)
   - Full-row-rank constraint matrix requirement: satisfied after Lemma 2.
   - Full-row-rank representation requirement for the composite regularizer (their Assumption 2 branch): satisfied after Lemma 3.
   - Feasible relative-interior initialization: assumed in part (ii) and preserved by Lemma 3(c).
   - Condition number: preserved exactly by Lemma 4.
   - Objective gap parameter ΔF_0: unchanged because F_0 and $x^{(0)}$ are unchanged.
   - Additional regularity appearing in their Assumption 1(e): I conservatively keep it as an explicit extra hypothesis of part (ii), since the accessible HTML suppresses the displayed formula at that point.

   Imported content. I do *not* re-prove the PG-RPD complexity theorem here; that part relies on the external proof in Liu–Lin–Xu (2025).

Self-confidence

3 — The counterexample and the reduction lemmas are fully checked; the only reason this is not a 4 is that the affirmative part intentionally relies on an audited but externally proved 2025 theorem, and one displayed regularity assumption in that source is partially suppressed by the accessible HTML.

Bibliography

```
\begin{verbatim}
@article{LiuLinXu2025Lower,
  author  = {Wei Liu and Qihang Lin and Yangyang Xu},
  title   = {Lower Complexity Bounds of First-order Methods for Affinely Constrained Composite Non-
    convex Problems},
  journal = {arXiv preprint arXiv:2502.17770},
  year    = {2025},
  doi     = {10.48550/arXiv.2502.17770},
  note    = {Accepted by Mathematics of Operations Research}
}

@article{LiuLinXu2025Near,
  author  = {Wei Liu and Qihang Lin and Yangyang Xu},
  title   = {A Near-optimal Method for Linearly Constrained Composite Non-convex Non-smooth
    Problems},
  journal = {arXiv preprint arXiv:2503.22927},
  year    = {2025},
  doi     = {10.48550/arXiv.2503.22927}
}

@article{LiuLinXu2023Extended,
  author  = {Wei Liu and Qihang Lin and Yangyang Xu},
  title   = {First-order Methods for Affinely Constrained Composite Non-convex Non-smooth Problems:
             Lower Complexity Bound and Near-optimal Methods},
  journal = {arXiv preprint arXiv:2307.07605},
  year    = {2023},
  doi     = {10.48550/arXiv.2307.07605}
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